Worked Example: CPP Compression via the ALCD

Robin Langer

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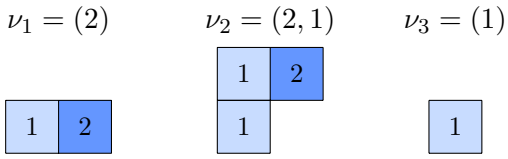
1 Setup

Profile $c = (2, 1, 0)$, $d = 3$. Consider the CPP $\lambda = \big((2),\ (2,1),\ (1)\big)$ with $\max(\lambda) = 2$ and $|\lambda| = 6$.

The interlacing chain is $\nu_1 \xrightarrow{1} \nu_2 \xrightarrow{0} \nu_3 \xrightarrow{2} \nu_1$.

2 The CPP with ring colouring

Each cell in column j of a Young diagram belongs to **ring** j. Peeling ring k demotes all parts equal to k (shortening rows of length k by one).



Numbers in cells indicate which ring they belong to. Light blue = ring 1, dark blue = ring 2.

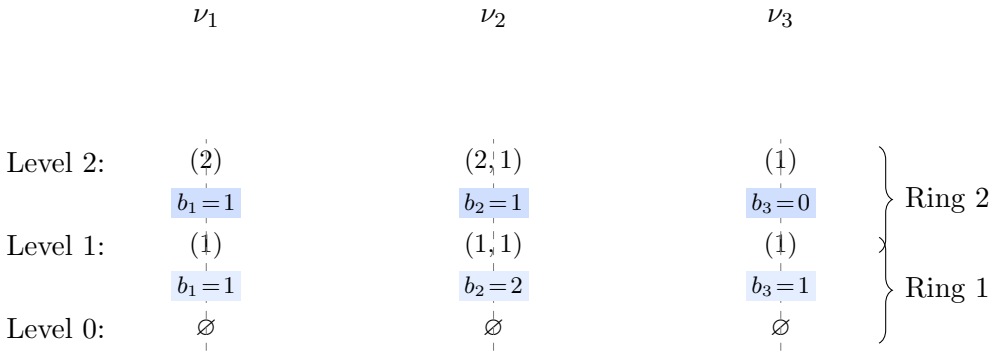
Face labels (= number of parts demoted per partition):

Ring k	b_1	b_2	b_3	total
$k = 2$	1	1	0	2
$k = 1$	1	2	1	4

Check: $2 + 4 = 6 = |\lambda|$.

3 The ALCD

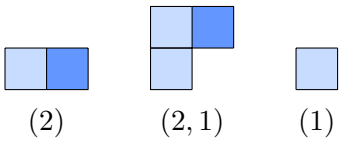
The ALCD (growth diagram) records the partitions at each level and the face labels between consecutive levels. Peeling proceeds top-down: ring 2 first, then ring 1.



4 Four compressions of λ

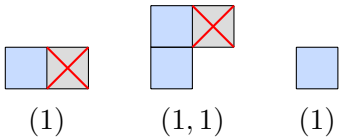
A *compression* selects a subset $S \subseteq \{1, \ldots, n\}$ of rings to **crush** (remove entirely). The remaining rings form a residual CPP μ with $\max(\mu) = n - |S|$. The sign is $(-1)^{|S|}$ and the total weight is $\sigma(S) + |\mu|$.

(a) $S = \varnothing$: keep all rings.



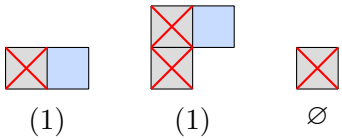
$\mu = \lambda$
sign $= +1$, $\quad |\mu| = 6$

(b) $S = \{2\}$: crush ring 2 (outermost).



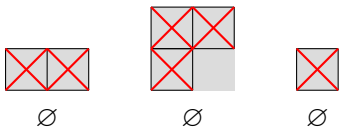
$\mu = \big((1), (1,1), (1)\big)$
sign $= -1$
$\sigma(S) + |\mu| = 2 + 4 = 6$

(c) $S = \{1\}$: crush ring 1 (innermost).



$\mu = \big((1), (1), \varnothing\big)$
sign $= -1$
$\sigma(S) + |\mu| = 1 + 2 = 3$

(d) $S = \{1, 2\}$: crush both rings.



$\mu = \big(\varnothing, \varnothing, \varnothing\big)$
sign $= +1$
$\sigma(S) + |\mu| = 3 + 0 = 3$

5 Cancellation

The four compressions of λ produce ground-set elements at two different total weights:

	S	μ	sign	total weight
(a)	$\varnothing$	λ	$+1$	6
(b)	$\{2\}$	$\big((1),(1,1),(1)\big)$	-1	6
(c)	$\{1\}$	$\big((1),(1),\varnothing\big)$	-1	3
(d)	$\{1,2\}$	$\big(\varnothing,\varnothing,\varnothing\big)$	$+1$	3

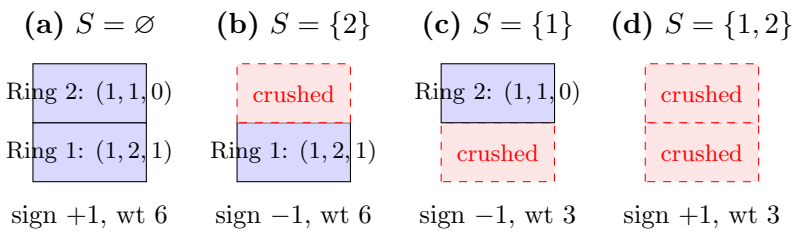
At weight 6: (a) has sign $+1$, (b) has sign -1. They cancel.
At weight 3: (d) has sign $+1$, (c) has sign -1. They cancel.

Every compression of λ cancels with a partner at the same total weight and opposite sign.

The conjecture $Q_n \in \mathbb{N}[q]$ asserts that this cancellation holds globally: across all CPPs and all compressions, the positive terms outweigh the negative ones at every power of q.

6 The ALCD viewpoint

The four compressions correspond to four ways of *slicing* the ALCD:



Crushing a ring changes the sign (by (-1)) and adds the ring index to the total weight (via $\sigma(S)$), while removing that ring's contribution $|\mu|$ decreases. The net effect: crushing ring k changes the total weight by $k -$ (weight of ring k).

Specifically:

- Crushing ring 2 changes weight by $+2 - 2 = 0$ (weight preserved).
- Crushing ring 1 changes weight by $+1 - 4 = -3$ (weight decreases by 3).

Compressions (a) and (b) are at the *same* weight because ring 2's face-label total equals its index. This is the key condition for a weight-preserving sign-reversing pairing.

7 Including U: the full ground set

The previous sections used $U = \varnothing$. The full ground-set element is a triple (S, U, μ) where:

- $S \subseteq \{1, \ldots, n\}$ — crushed rings (from the Euler product $(zq;q)_\infty$).
- μ — residual CPP with $\max(\mu) = m := n - |S|$.
- $U \subseteq \{1, \ldots, m\}$ — **marked rings** of μ (from the Pochhammer $(q)_m$).

The sign is $(-1)^{|S|+|U|}$ and the total weight is $\sigma(S) + \sigma(U) + |\mu|$.

The Pochhammer factor $(q)_m = \prod_{k=1}^{m}(1 - q^k)$ expands as

$$(q)_m = \sum_{U \subseteq \{1,\ldots,m\}} (-1)^{|U|}\, q^{\sigma(U)}.$$

Each subset U "marks" certain rings of μ. Marking ring k flips the sign and adds k to the weight.

Full table from $\lambda = \big((2),(2,1),(1)\big)$

With U included, a single CPP generates many more ground-set elements:

	S	U	μ	sign	total wt
1	$\varnothing$	$\varnothing$	λ	$+$	6
2	$\{2\}$	$\varnothing$	$\big((1),(1,1),(1)\big)$	$-$	6
3	$\varnothing$	$\{1\}$	λ	$-$	7
4	$\{2\}$	$\{1\}$	$\big((1),(1,1),(1)\big)$	$+$	7
5	$\varnothing$	$\{2\}$	λ	$-$	8
6	$\varnothing$	$\{1,2\}$	λ	$+$	9
7	$\{1\}$	$\varnothing$	$\big((1),(1),\varnothing\big)$	$-$	3
8	$\{1,2\}$	$\varnothing$	$\big(\varnothing,\varnothing,\varnothing\big)$	$+$	3
9	$\{1\}$	$\{1\}$	$\big((1),(1),\varnothing\big)$	$+$	4

Cancellation within this CPP:

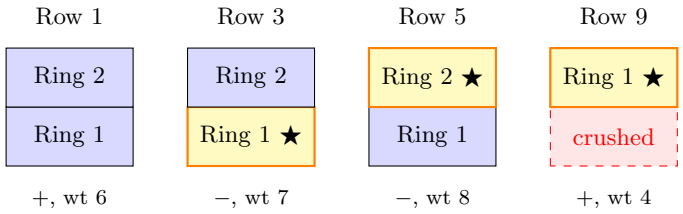
- Rows 1 & 2 cancel at weight 6 ($S = \varnothing$ vs $S = \{2\}$, same $U = \varnothing$).
- Rows 3 & 4 cancel at weight 7 ($S = \varnothing$ vs $S = \{2\}$, same $U = \{1\}$).
- Rows 7 & 8 cancel at weight 3 ($S = \{1\}$ vs $S = \{1,2\}$, same $U = \varnothing$).

Survivors from this CPP:

- Row 5: sign $-$, weight 8. Needs a partner from a *different* CPP.
- Row 6: sign $+$, weight 9. Needs a partner from a different CPP.
- Row 9: sign $+$, weight 4. Needs a partner from a different CPP.

ALCD with U-marking

Visually, U marks rings *within* the residual μ. Marked rings are shown with a star.



Three operations on the ALCD, each flipping the sign:

1. **Crush** a ring (move index from μ to S): adds k to $\sigma(S)$, removes ring k's weight from $|\mu|$.
2. **Mark** a ring (add index to U): adds k to $\sigma(U)$, keeps $|\mu|$ unchanged.
3. **Unmark** a ring (remove index from U): subtracts k from $\sigma(U)$.

The involution ι must choose one of these operations at each step, always preserving the total weight $\sigma(S) + \sigma(U) + |\mu|$.